Lab 12: Algorithms with AI Assistance – Sorting, Searching, and Optimizing Algorithms

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Task Description #1 (Sorting – Merge Sort Implementation)
Prompt: Use AI to generate a Python program that implements the Merge Sort algorithm.
Input:

```
def merge_sort(arr):
    """
    Sort a list in ascending order using Merge Sort.

    Time Complexity: O(n log n)
    Space Complexity: O(n)

    Args:
        arr: A list of elements to be sorted.

    Returns:
        A new sorted list in ascending order.
```

```
    """
    if len(arr) <= 1:
        return arr.copy()

    mid = len(arr) // 2

    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    return merge(left, right)


def merge(left, right):
    """Merge two sorted lists into one sorted list."""
    result = []
    i = 0
    j = 0

    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
```

```
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])

    return result



# Test cases
numbers = [38, 27, 43, 3, 9, 82, 10]

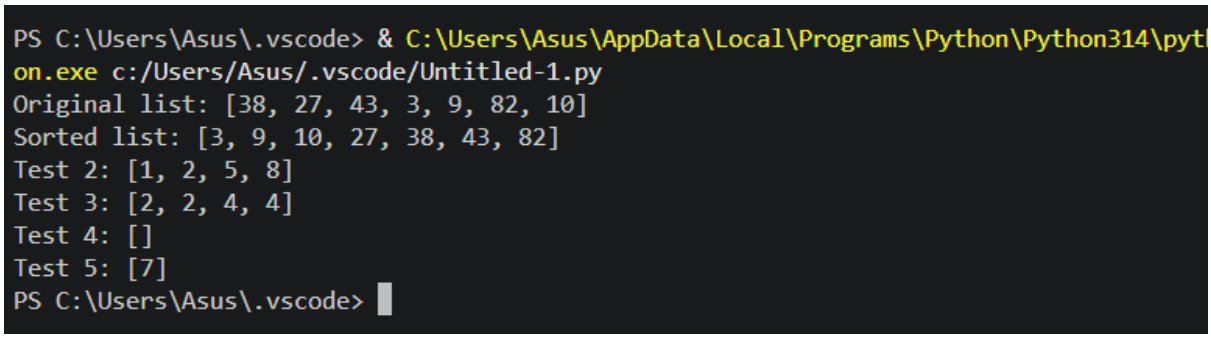
print("Original list:", numbers)
print("Sorted list:", merge_sort(numbers))

print("Test 2:", merge_sort([5, 2, 8, 1]))
print("Test 3:", merge_sort([4, 4, 2, 2]))
print("Test 4:", merge_sort([]))
print("Test 5:", merge_sort([7]))
```

Explanation:

Merge Sort is a sorting algorithm that arranges elements in ascending or descending order. It uses the Divide and Conquer technique, which means dividing a problem into smaller parts, solving them, and combining the results.

Output:



Task Description #2 (Searching – Binary Search with AI Optimization)

Prompt: Use AI to create a binary search function that finds a target element in a sorted list.

Input:

```
def binary_search(arr, target):
    """
    Search for a target element in a sorted list.

    Args:
        arr: A list sorted in ascending order.
        target: The element to search for.

    Returns:
        The index of the target if found, otherwise -1.
```

```
Time Complexity:
    Best case: O(1)
    Average case: O(log n)
    Worst case: O(log n)

Space Complexity: O(1)
"""
low = 0
high = len(arr) - 1

# Continue searching while the range is valid
while low <= high:
    # Find the middle index
    mid = (low + high) // 2

    # Check whether the middle element is the target
    if arr[mid] == target:
        return mid

    # Search the right half if target is greater
    elif arr[mid] < target:
        low = mid + 1

    # Search the left half if target is smaller
```

```
        else:
            high = mid - 1

    # Return -1 if the target is not found
    return -1



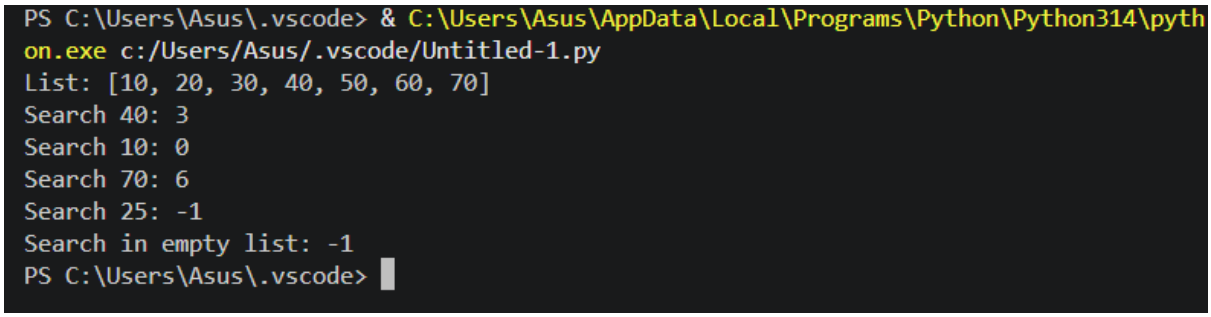
# Test cases
numbers = [10, 20, 30, 40, 50, 60, 70]

print("List:", numbers)
print("Search 40:", binary_search(numbers, 40))
print("Search 10:", binary_search(numbers, 10))
print("Search 70:", binary_search(numbers, 70))
print("Search 25:", binary_search(numbers, 25))
print("Search in empty list:", binary_search([], 10))
```

Explanation:

Binary Search finds a target element in a sorted list by repeatedly dividing the search range into two halves, returning its index if found or -1 otherwise.

Output:



Task Description #3 (Real-Time Application – Inventory Management System)

 Scenario: A retail store's inventory system contains thousands of products, each with attributes like product ID, name, price, and stock quantity. Store staff need to:

1. Quickly search for a product by ID or name.
2. Sort products by price or quantity for stock analysis.

Prompt:

o Use AI to suggest the most efficient search and sort algorithms for this use case.

o Implement the recommended algorithms in Python.

o Justify the choice based on dataset size, update frequency, and performance requirements.

Input:

```
# Inventory Management System

inventory = [
    {"id": 101, "name": "Laptop", "price": 55000, "quantity": 10},
```

```
    {"id": 102, "name": "Mouse", "price": 500, "quantity": 50},
    {"id": 103, "name": "Keyboard", "price": 1200, "quantity": 30},
    {"id": 104, "name": "Monitor", "price": 12000, "quantity": 15},
    {"id": 105, "name": "Headphones", "price": 2000, "quantity": 25}
]

# Create a dictionary for fast product ID searching
product_index = {product["id"]: product for product in inventory}


def search_by_id(product_id):
    """Search by ID. Average time complexity: O(1)."""
    return product_index.get(product_id, "Product not found")


def search_by_name(name):
    """Search by name. Time complexity: O(n)."""
    for product in inventory:
        if product["name"].lower() == name.lower():
            return product
    return "Product not found"


def merge_sort(products, key):
```

```
    """
    Sort products in ascending order using Merge Sort.
    Time complexity: O(n log n).
    Auxiliary space complexity: O(n).
    """
    if len(products) <= 1:
        return products.copy()

    mid = len(products) // 2
    left = merge_sort(products[:mid], key)
    right = merge_sort(products[mid:], key)

    return merge(left, right, key)



def merge(left, right, key):
    """Merge two sorted product lists."""
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
            i += 1
```

```
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


def display(products):
    """Display product details."""
    for product in products:
        print(product)


# Search operations
print("Search by ID:")
print(search_by_id(103))

print("\nSearch by Name:")
print(search_by_name("Laptop"))

# Sorting operations
print("\nProducts sorted by price:")
```

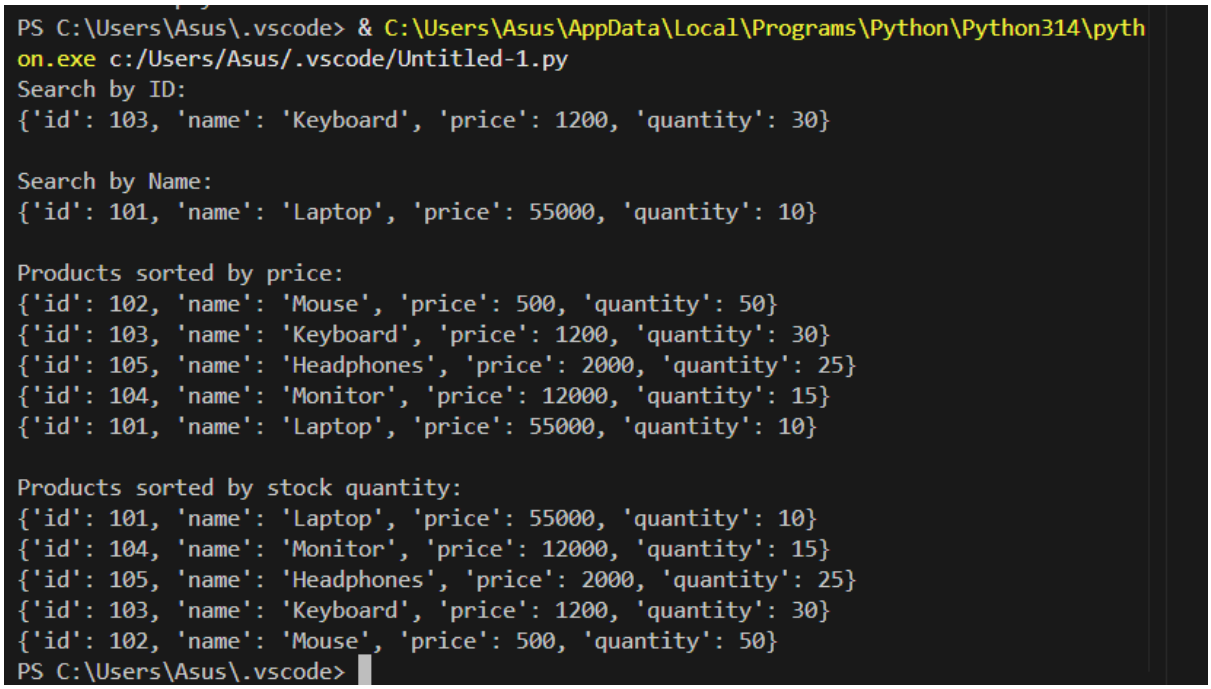
```
display(merge_sort(inventory, "price"))


print("\nProducts sorted by stock quantity:")
display(merge_sort(inventory, "quantity"))
```

Explanation:

The Inventory Management System uses a dictionary for fast product ID searching, linear search for product names, and Merge Sort to arrange products by price or stock quantity efficiently.

Output:



Task description #4: Smart Hospital Patient Management System

A hospital maintains records of thousands of patients with

details

such as patient ID, name, severity level, admission date, and bill amount. Doctors and staff need to:

1. Quickly search patient records using patient ID or name.
2. Sort patients based on severity level or bill amount for prioritization and billing.

Student Task

 Use AI to recommend suitable searching and sorting algorithms.

 Justify the selected algorithms in terms of efficiency and suitability.

 Implement the recommended algorithms in Python.

Input:

```
patients = [
    {"id": 101, "name": "Ravi", "severity": 2, "admission_date": "2026-10-10", "bill": 15000},
    {"id": 102, "name": "Anjali", "severity": 1, "admission_date": "2026-10-11", "bill": 25000},
    {"id": 103, "name": "Kiran", "severity": 3, "admission_date": "2026-10-12", "bill": 10000},
    {"id": 104, "name": "Priya", "severity": 2, "admission_date": "2026-10-13", "bill": 30000}
]
```

```
# Fast patient ID search using a dictionary
patient_index = {p["id"]: p for p in patients}

def search_by_id(patient_id):
    return patient_index.get(patient_id, "Patient not found")

# Patient name search using Linear Search
def search_by_name(name):
    for patient in patients:
        if patient["name"].lower() == name.lower():
            return patient
    return "Patient not found"

# Merge Sort
def merge_sort(data, key):
    if len(data) <= 1:
        return data.copy()

    mid = len(data) // 2
    left = merge_sort(data[:mid], key)
    right = merge_sort(data[mid:], key)

    return merge(left, right, key)
```

```
def merge(left, right, key):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

# Display patient records
print("Search by Patient ID:")
print(search_by_id(102))

print("\nSearch by Patient Name:")
print(search_by_name("Ravi"))
```

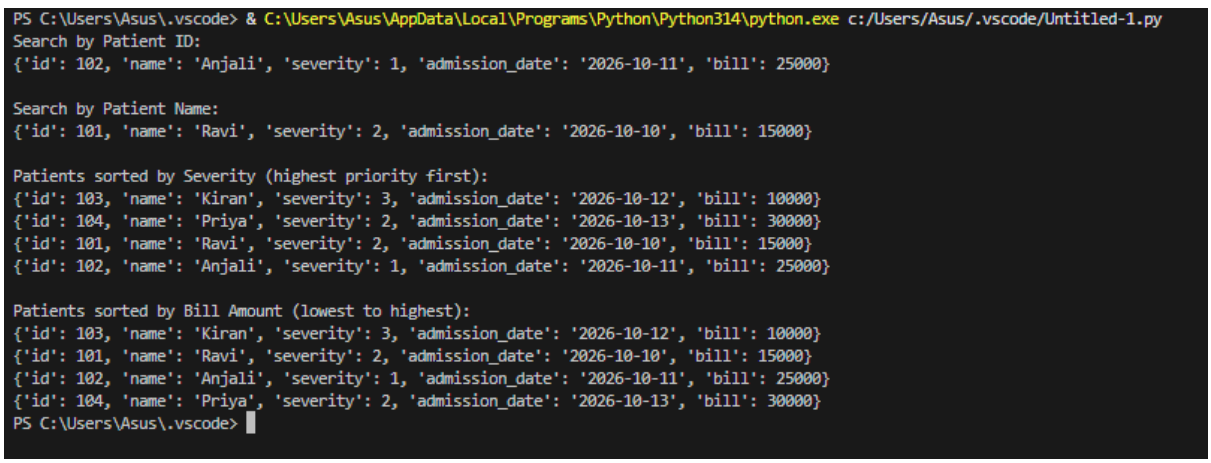
```
print("\nPatients sorted by Severity (highest priority first):")
display_severity = merge_sort(patients, "severity")
display_severity.reverse()
for patient in display_severity:
    print(patient)

print("\nPatients sorted by Bill Amount (lowest to highest):")
for patient in merge_sort(patients, "bill"):
    print(patient)
```

Explanation:

The Smart Hospital Patient Management System uses a dictionary for fast patient ID searches, linear search for patient names, and Merge Sort to arrange patients by severity level or bill amount efficiently.

Output:



Task Description #5: University Examination Result Processing System

A university processes examination results for thousands of students

containing roll number, name, subject, and marks. The system must:

1. Search student results using roll number.

2. Sort students based on marks to generate rank lists.

Student Task

 Identify efficient searching and sorting algorithms using AI

assistance.

 Justify the choice of algorithms.

 Implement the algorithms in Python.

Input:

```
students = [
    {"roll_no": 101, "name": "Anjali", "subject": "Python", "marks": 85},
    {"roll_no": 102, "name": "Ravi", "subject": "Python", "marks": 92},
    {"roll_no": 103, "name": "Priya", "subject": "Python", "marks": 78},
    {"roll_no": 104, "name": "Kiran", "subject": "Python", "marks": 95}
]

# Search using Dictionary
student_index = {s["roll_no"]: s for s in students}
```

```
def search_student(roll_no):
    return student_index.get(roll_no, "Student not found")

# Sort using Merge Sort
def merge_sort(data):
    if len(data) <= 1:
        return data.copy()

    mid = len(data) // 2
    left = merge_sort(data[:mid])
    right = merge_sort(data[mid:])

    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i]["marks"] >= right[j]["marks"]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
```

```
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

print("Search Result:")
print(search_student(102))

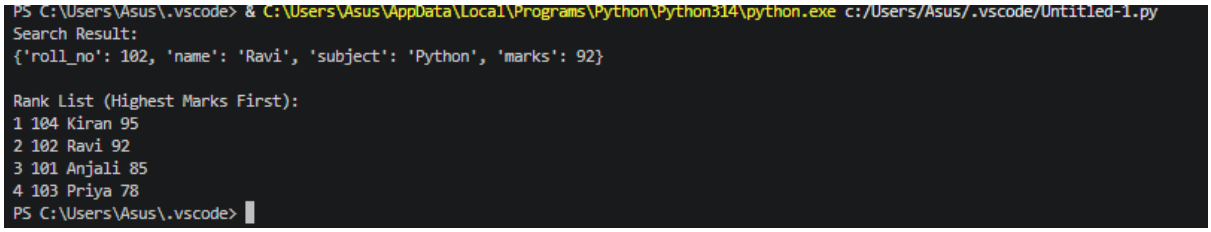
print("\nRank List (Highest Marks First):")
rank_list = merge_sort(students)

for rank, student in enumerate(rank_list, start=1):
    print(rank, student["roll_no"], student["name"],
student["marks"])
```

Explanation:

The system uses a dictionary for fast roll-number searching and Merge Sort to generate a rank list from highest to lowest marks, with average search time \(O(1)\) and sorting time \(O(n\log n)\).

Output:



Task Description #6: Online Food Delivery Platform

An online food delivery application stores thousands of orders with

order ID, restaurant name, delivery time, price, and order status. The

platform needs to:

1. Quickly find an order using order ID.

2. Sort orders based on delivery time or price.

Student Task

 Use AI to suggest optimized algorithms.

 Justify the algorithm selection.

 Implement searching and sorting modules in Python.

Input:

```
orders = [
    {"order_id": 101, "restaurant": "Dominos", "delivery_time": 30, "price": 500, "status": "Delivered"},
    {"order_id": 102, "restaurant": "Burger King", "delivery_time": 20, "price": 300, "status": "Preparing"},
    {"order_id": 103, "restaurant": "Pizza Hut", "delivery_time": 40, "price": 700, "status": "On the way"},
    {"order_id": 104, "restaurant": "KFC", "delivery_time": 25, "price": 400, "status": "Delivered"}
]
```

```
# Fast search using Dictionary
order_index = {order["order_id"]: order for order in orders}

def search_order(order_id):
    return order_index.get(order_id, "Order not found")

# Merge Sort
def merge_sort(data, key):
    if len(data) <= 1:
        return data.copy()

    mid = len(data) // 2
    left = merge_sort(data[:mid], key)
    right = merge_sort(data[mid:], key)

    return merge(left, right, key)

def merge(left, right, key):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
```

```
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

# Search an order
print("Search Order:")
print(search_order(102))

# Sort by delivery time
print("\nOrders sorted by delivery time:")
for order in merge_sort(orders, "delivery_time"):
    print(order)

# Sort by price
print("\nOrders sorted by price:")
for order in merge_sort(orders, "price"):
    print(order)
```

Explanation:

The system uses a dictionary to find orders quickly by order ID

and Merge Sort to arrange orders by delivery time or price.

Output:

